

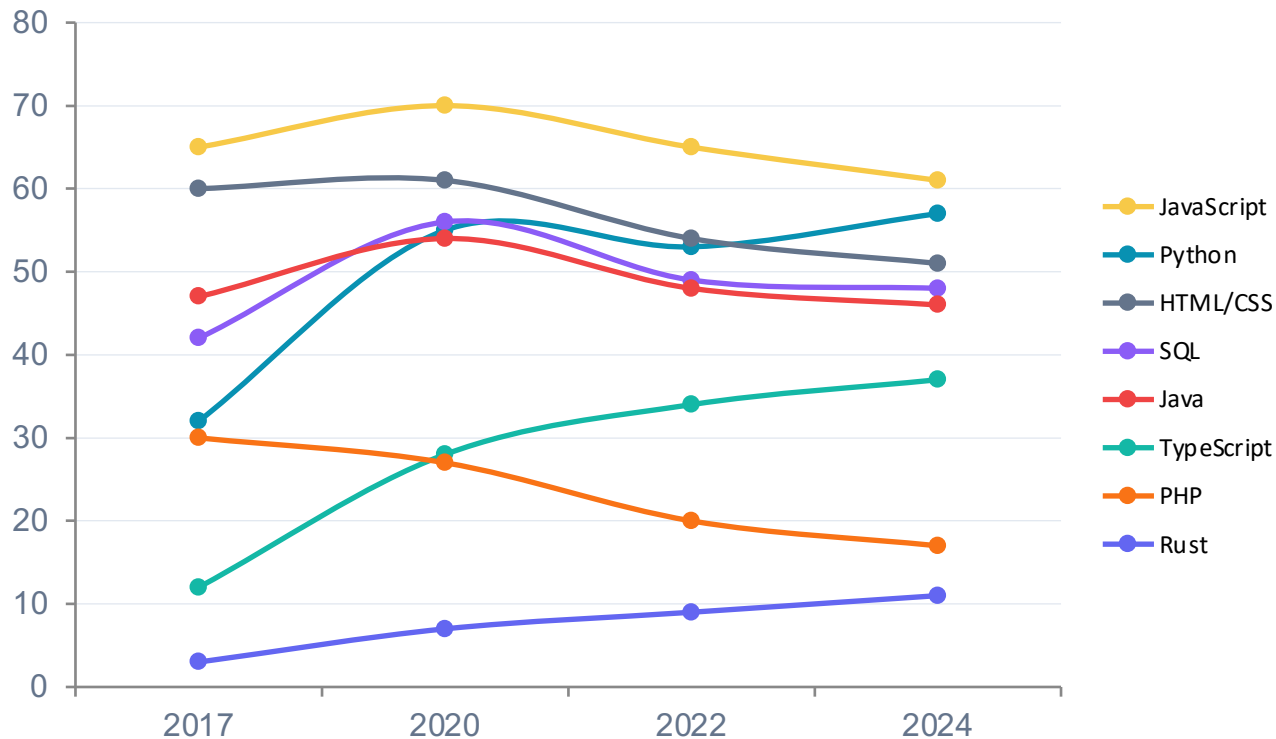
State of Developer Ecosystem 2024

Language Trends, Satisfaction, and Strategic Shifts

Based on JetBrains survey of 23,262 developers worldwide

For Engineering Leaders, CTOs & Developer Experience Teams

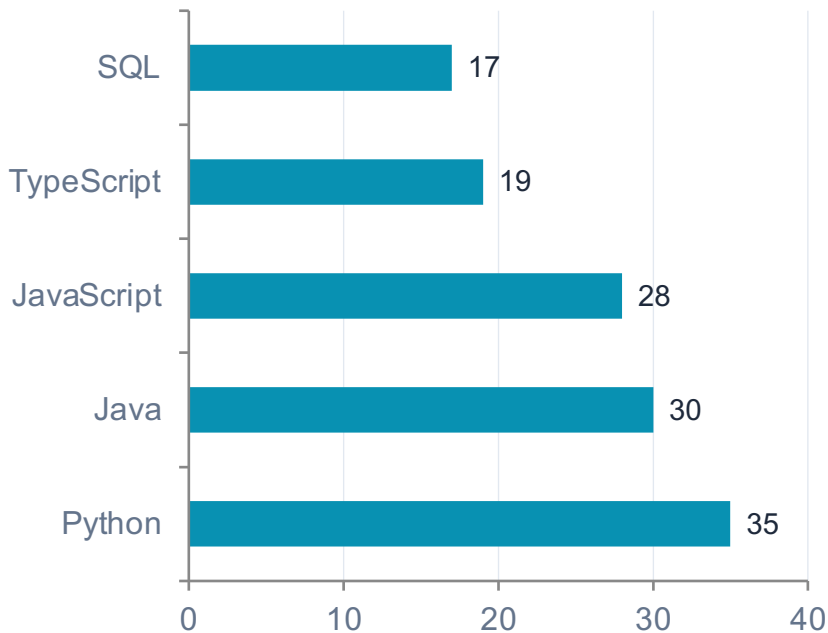
Python Closes the Gap: 8 Years of Language Usage Shifts



Key Shifts

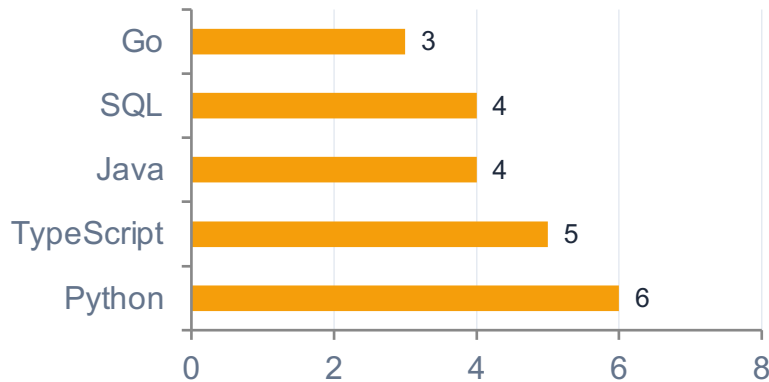
- JavaScript peaked at 70% (2020), now 61%
- Python rose from 32% to 57%
- TypeScript tripled: 12% → 37%
- PHP dropped from 30% to 17%
- Rust grew from ~0 to 11%

Python Is Now the #1 Primary Language at 35%



Adoption Intent

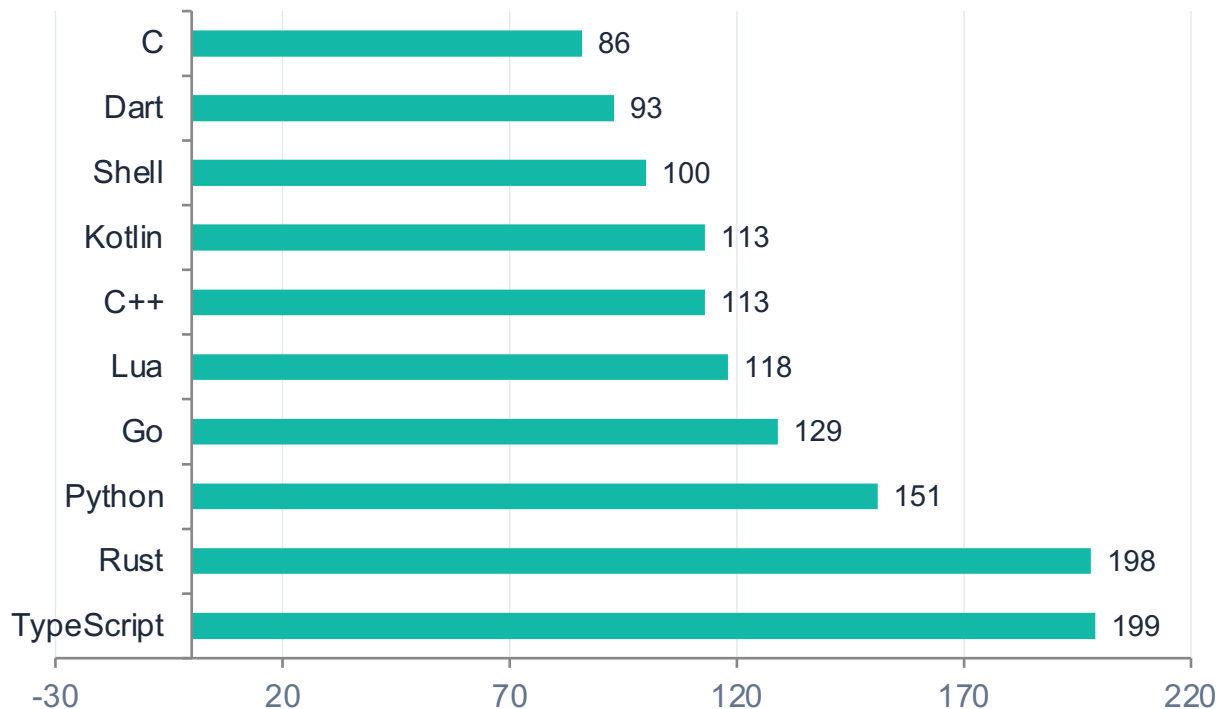
Only 4–6% of developers plan to adopt any new language



Ecosystem lock-in effect: Developers rarely switch primary languages, despite rising stars like Rust and Go

TypeScript and Rust Top Developer Satisfaction

Nearly tied at 199 vs 198 — reflecting enthusiasm for type safety and modern tooling



Why TypeScript & Rust?

- Type safety prevents bugs
- Modern tooling & DX
- Strong type systems
- Zero-cost abstractions (Rust)
- Memory safety guarantees

Each Language Owns a Domain

AI belongs to Python, UI to TypeScript — language choice is driven by domain, not general preference

Domain	Leader	Share	Key Advantage
AI / Machine Learning	Python	31%	Rich ML ecosystem, TensorFlow, PyTorch
User Interface	TypeScript	62%	React, Vue, Angular frameworks
APIs & Services	Java	49%	Spring Boot, mature enterprise stack
System Tools & Components	C++	27%	Performance, hardware-level control
Data Science & Analytics	Python	28%	Pandas, NumPy, Jupyter notebooks
Data Processing	Python	17%	Pandas, Spark, ETL pipelines

What This Means for Your Technology Strategy in 2025

1 Invest in Python for AI/ML

Undisputed leader at 31% in AI/ML. Teams need Python expertise.

2 TypeScript is the future of frontend

62% UI share, 37% overall. TypeScript skills are critical for modern web.

3 Rust & Go are growth plays

Highest satisfaction scores and rising usage. Ideal for infrastructure.

4 PHP & Ruby pools shrinking

PHP dropped 30%→17% since 2017. Ruby fell 10%→4%. Plan accordingly.

5 Satisfaction = type safety + DX

TypeScript & Rust top satisfaction. Factor into developer retention.